Sample Notes: Elementary Theorem Environments

November 2, 2025

1 Integers and Parity

In these notes we demonstrate the environments using elementary concepts. We refer to Definition 1.1, Lemma 1.1, Theorem 1.5, Proposition 1.7, Corollary 1.9, Claim 1.3, and Exercise 1.1.

Definition 1.1

An integer n is *even* if there exists $k \in \mathbb{Z}$ with $n = 2k$. An integer is *odd* if there exists $k \in \mathbb{Z}$ with $n = 2k + 1$.

Example 1.1.1

The integers 4, -6, and 0 are even; 3 and 7 are odd.

Example 1.1.2

Since $12 = 2 \cdot 6$ and $-8 = 2 \cdot (-4)$, both 12 and -8 are even. Conversely, 5 is not even because it cannot be written as $2k$ for any integer k.

Lemma 1.1

If n is even, then n^2 is even.

Proof 1.2

Write $n = 2k$ for some $k \in \mathbb{Z}$. Then $n^2 = (2k)^2 = 4k^2 = 2\,(2k^2)$, which is even.

Claim 1.3

The sum of two odd integers is even.

Proof 1.4

Let $a = 2r + 1$ and $b = 2s + 1$ with $r, s \in \mathbb{Z}$. Then $a + b = 2(r + s + 1)$ is even.

Theorem 1.5

For integers a and b:

- even + even = even,
- even + odd = odd,
- odd + odd = even.

Proof 1.6

Write a and b as $2r$ or $2r+1$ and add; each case follows immediately. The final case is exactly Claim 1.3.

Proposition 1.7

If $a \mid b$ and $b \mid c$ for integers a, b, c, then $a \mid c$.

Proof 1.8

There exist $m, n \in \mathbb{Z}$ with $b = am$ and $c = bn$. Then $c = (am)n = a(mn)$, so $a \mid c$.

Corollary 1.9

If n is even, then n^2 is divisible by 4.

Proof 1.10

By Lemma 1.1 write $n = 2k$. Then $n^2 = 4k^2$, a multiple of 4.

Remark. From Definition 1.1, every integer is exactly one of even or odd.

Exercise 1.1

Show that the product of two odd integers is odd.

Solution

Let $a = 2r+1$ and $b = 2s+1$. Then

$$ab = (2r+1)(2s+1) = 4rs + 2r + 2s + 1 = 2\big(2rs + r + s\big) + 1,$$

which is odd.